by virtue of its properties of setting or hardening in combination with water. It helps to fill the voids and imparts density to concrete.

4.3.3.2 Fine Aggregate

Sand or crushed stone sand (M sand) are used as fine aggregate to fill the voids leaved between coarse aggregates and thereby reduce the quantity of cement.

4.3.3.3 Coarse Aggregate

Broken stone or broken brick acts as main filler and forms the main bulk of concrete. The aggregates should be clean, dense, hard, strong and durable.

4.3.3.4 Water

Water facilitates the spreading of cement over the aggregates and regulates the consistency. Water used should be clean. Sea water should not be used as it retards setting of concrete.

4.3.3.5 Admixtures in cement

There are also certain other additives known as "**Admixtures**" added to improve the quality of concrete required for various constructions. Some of them are,

i) Plasticizers.
ii) Superplasticizers.
iii) Accelerators.
iv) Retarders.
v) Pozzolanic material.
vi) Air entraining agents.
vii) Fibres.
viii) Polymers.
ix) Silica fume.



Plasticizers



Superplasticizers



Accelerators

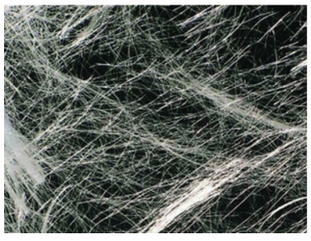


Retarders


Pozzolanic material



Air entraining agents


Fibres


Polymers



Silica fume

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